AI-POWERED CUSTOMER SUPPORT TICKETING SYSTEM

Project documentation & implementation evidence

Salesforce Service Cloud • Record-Triggered Flows • SLA Monitoring • Agentforce Configuration

Project focus

Centralize support cases, classify and prioritize incidents, track resolution targets, support agents with suggested actions, and monitor outcomes through feedback and reporting.

Scope: Salesforce DX metadata package covering custom objects and Case fields, record-triggered Flows, validation, email templates and alerts, support profiles, reports, a Lightning app, and Agentforce-related

configuration.

> **Verification boundary.** This document distinguishes packaged metadata, results reported from the live development org, and items that remain unverified. Salesforce CLI deployment validation has not been performed; Agentforce's intended troubleshooting workflow has not been proven end-to-end.

Prepared from the supplied project package, existing implementation notes, and screenshots captured from the supplied demonstration video. Screenshots illustrate visible UI states; they do not independently prove every backend behavior.

1. Executive Summary

The Salesforce AI-Powered Customer Support Ticketing System is designed for common telecommunications support scenarios, including broadband connectivity outages, eSIM activation failures, roaming billing disputes, and network incidents. It brings case records, priority-based service targets, automation, agent-facing suggested replies, feedback capture, and operational reporting into a single Salesforce workspace.

Business problem

Support teams need a consistent way to classify incoming incidents, assign priority, track expected resolution time, recognize overdue cases, and review the quality of agent assistance. When these steps are manual or spread across separate records, triage becomes inconsistent and SLA risks are harder to monitor.

Solution approach

The package extends the standard Case object and adds custom objects for support categories, SLA policies, and AI feedback. Record-triggered Flows assign selected Case values and drive notification paths. Reports provide operational views of open cases and SLA status. Agentforce-related metadata is included, but successful custom troubleshooting action execution remains unverified.

Key implementation components

Area	Implementation in package
Data model	Case plus Support_Category__c, SLA_Policy__c, AI_Feedback__c
Automation	Before-save and after-save record-triggered Flows; resolution notification Flow
Controls	High-priority SLA due-date validation rule; support profiles
Agent support	AI_Suggested_Reply__c field and Agentforce-related metadata
Monitoring	Open Cases by Priority and SLA Breach & Resolution Time Analysis reports

Configured SLA targets

Case priority	Target resolution time
High	4 hours
Medium	12 hours
Low	24 hours

2. Requirements and Functional Scope

The functional scope below is derived from the project definition and the packaged metadata. Where a behavior depends on runtime execution, the status is stated separately rather than assumed from the existence of configuration files.

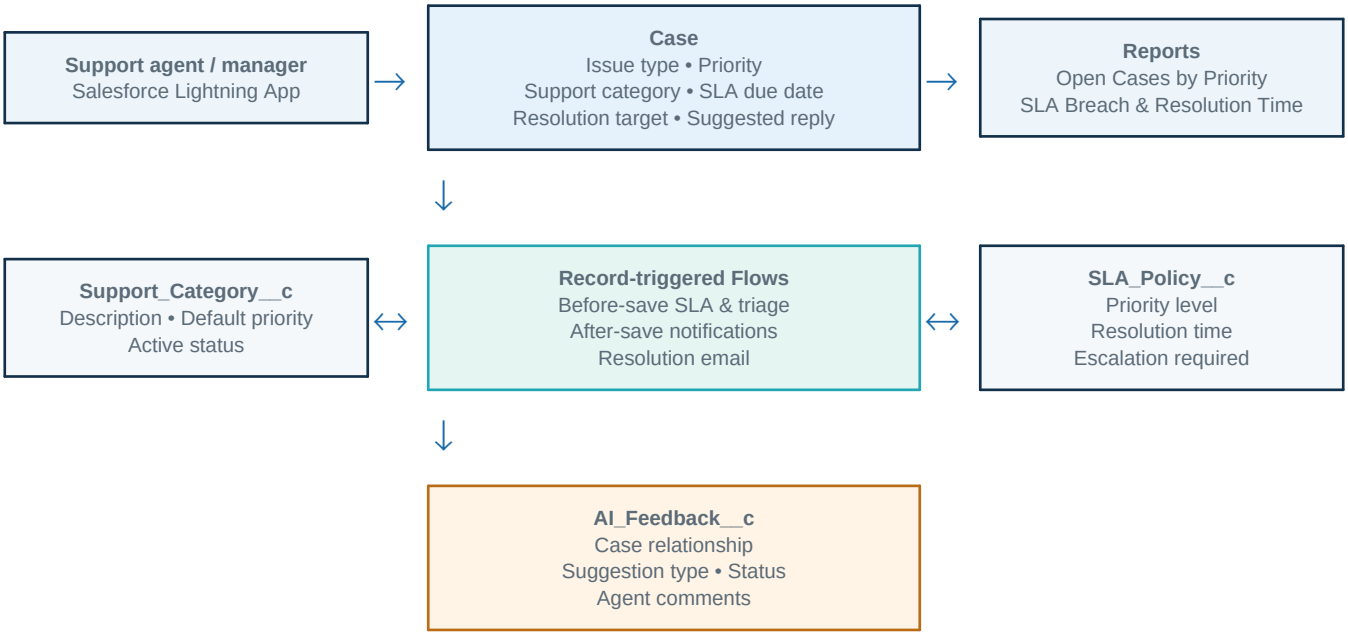
Requirement area	Intended behavior	Evidence boundary
Case management	Manage customer support incidents using Salesforce Case records.	Case records are visible in the supplied demo.
Classification & priority	Use issue type, support category, and priority fields to structure triage.	Flow metadata is present; verify in a target org before operational use.
SLA tracking	Populate a resolution target and SLA due date based on configured logic.	Fields, Flow metadata, and reports are packaged; combined behavior requires target-org tests.
Agent assistance	Expose a suggested-reply field and Agentforce-related configuration.	The preview returned a fallback escalation response; successful custom troubleshooting was not demonstrated.
Feedback loop	Link AI feedback records to Cases and capture feedback status/type/comments.	Two feedback records were shown in the supplied demonstration; IDs are environment-specific.
Monitoring	Review open cases by priority and SLA breach/resolution-time status.	Both report definitions are included in the package.
Access control	Provide Support Agent and Support Manager profiles.	Profiles are packaged; review permissions in the target org.
Maintenance	Review Flow errors, data quality, report trends, and access configuration.	Operational procedures are recommended; no continuous monitoring service is claimed.

Stakeholders

Stakeholder	Primary concern
Support agent	Clear case details, priority, SLA deadline, and usable suggested actions.
Support manager	Escalated/overdue cases, workload visibility, and support performance.
Salesforce administrator	Flow health, field configuration, permissions, deployment, and troubleshooting.
Project reviewer	Evidence that documented behavior matches the packaged implementation and demo.

3. Architecture and Data Model

The following conceptual architecture summarizes how the packaged components relate. It is a documentation diagram, not a generated Salesforce schema export.



Packaged custom objects

Object	Purpose and notable data
Support_Category__c	Categorizes incidents; includes Description, Default_Priority__c, and Active__c.
SLA_Policy__c	Stores Priority_Level__c, Resolution_Time__c, and Escalation_Required__c.
AI_Feedback__c	Stores a relationship to Case plus AI_Suggestion_Type__c, Feedback_Status__c, and Agent_Comments__c.
Case (standard)	Extended with Issue_Type__c, Support_Category__c, SLA_Due_Date__c, Resolution_Time__c, and AI_Suggested_Reply__c.

Relationships and data handling

AI_Feedback__c links feedback to a Case. The Case's Support_Category__c field connects case triage to the support-category configuration. SLA policy metadata stores resolution targets and escalation settings. Confirm the actual field relationship types and lookup behavior in the target org before migration; this document does not infer unlisted relationships.

4. Automation Logic and Validation

Component	Configured responsibility	Caution / verification
Case_Before_Save_SLA_and_Triage_Flow	Before-save field assignments for SLA target/deadline and suggested reply, under its configured conditions.	Test inserts and updates for High, Medium, and Low priorities; verify entry conditions and null handling.
Case_SLA_and_Triage_Automation_Flow	After-save notification paths for High-priority assignment and transition to Escalated.	Confirm recipient, activation, email deliverability, and duplicate-notification behavior.
Case_Resolution_Email_Flow	Sends a resolution notification when a Case transitions to Closed.	Test status transition and email content in a disposable org.
Case_Summary_Flow	Autolaunched Flow referenced by Agentforce-related metadata.	Successful live invocation was not established by the preview test.

High-priority SLA validation rule

Name: SLA_Due_Date_Required_For_High_Priority
Formula: AND(ISPICKVAL(Priority, "High"), ISBLANK(SLA_Due_Date__c))
Error: “SLA Due Date must be populated for High priority cases.”

The rule is intended to prevent saving a High-priority Case without an SLA due date. Test both negative and positive paths: a High-priority Case with a blank deadline should be blocked, while a valid deadline should allow saving. Also test whether the Flow populates the field before validation in the actual execution context.

Operational troubleshooting procedure

Step	Administrator action
1. Reproduce	Record the affected Case, field values, user context, and exact operation that failed.
2. Inspect automation	Review Flow error emails, failed Flow interviews/debug details, entry conditions, and element outcomes.
3. Inspect data	Check required fields, picklist values, record relationships, and record history.
4. Inspect access	Review profile and permission-set object/field permissions for the affected user.
5. Correct and retest	Apply the smallest justified configuration change and rerun both positive and negative test cases.
6. Monitor	Review case volume, SLA report classifications, notification outcomes, and feedback records after the fix.

5. Demonstration Evidence

The following images are frames extracted from the supplied final demonstration video. They illustrate the visible Salesforce interface at the indicated points. They are visual evidence of the displayed screens, not independent proof of deployment, security enforcement, or end-to-end automation.

Salesforce setup and escalation configuration

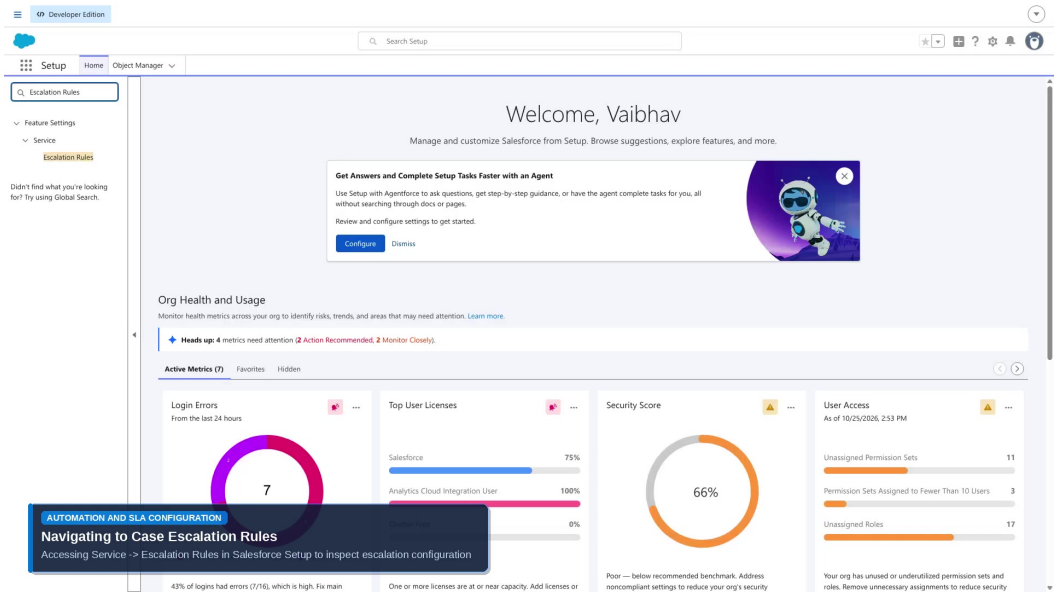


Figure 1. Setup / escalation-rule area shown during the automation and SLA configuration section (approximately 02:52).

AI Feedback records and reporting

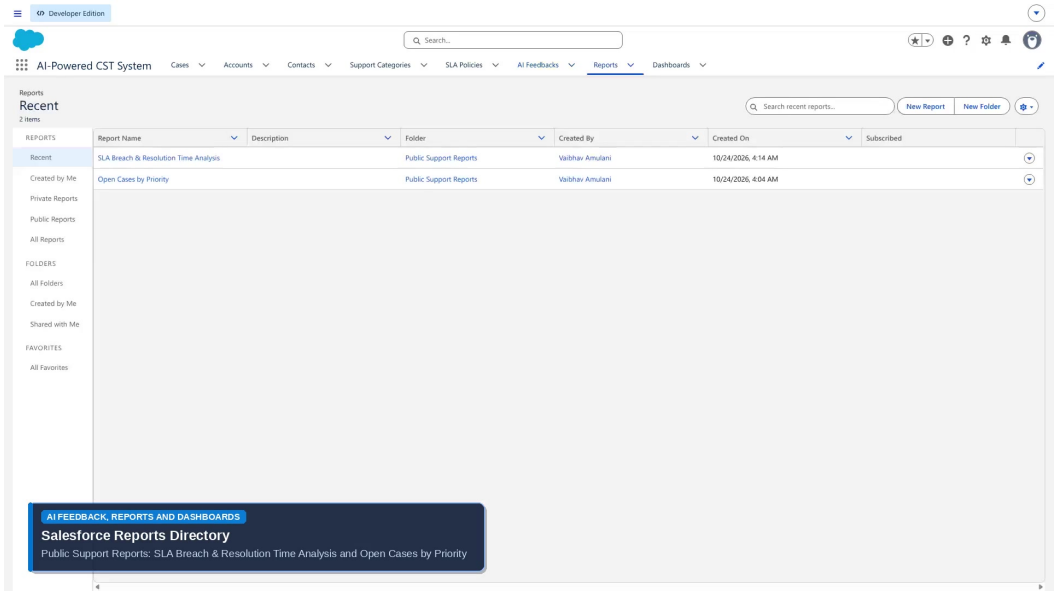


Figure 2. Salesforce report interface shown during the AI Feedback, Reports & Dashboards section (approximately 04:11).

5. Demonstration Evidence (continued)

Agentforce configuration

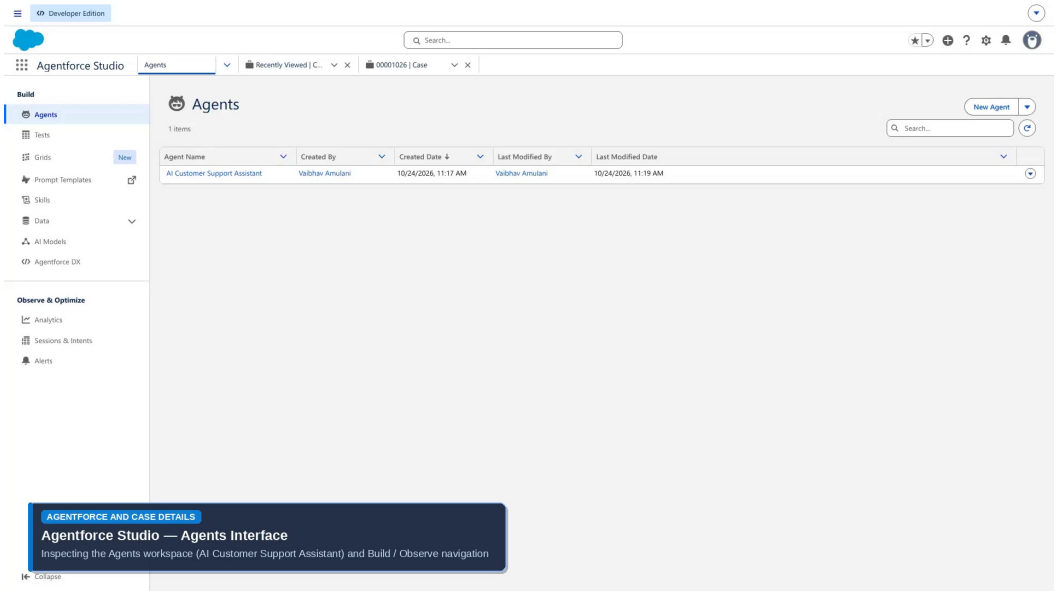


Figure 3. Agentforce Studio / Agents interface visible during the Agentforce section (approximately 04:49).

Case details and related records

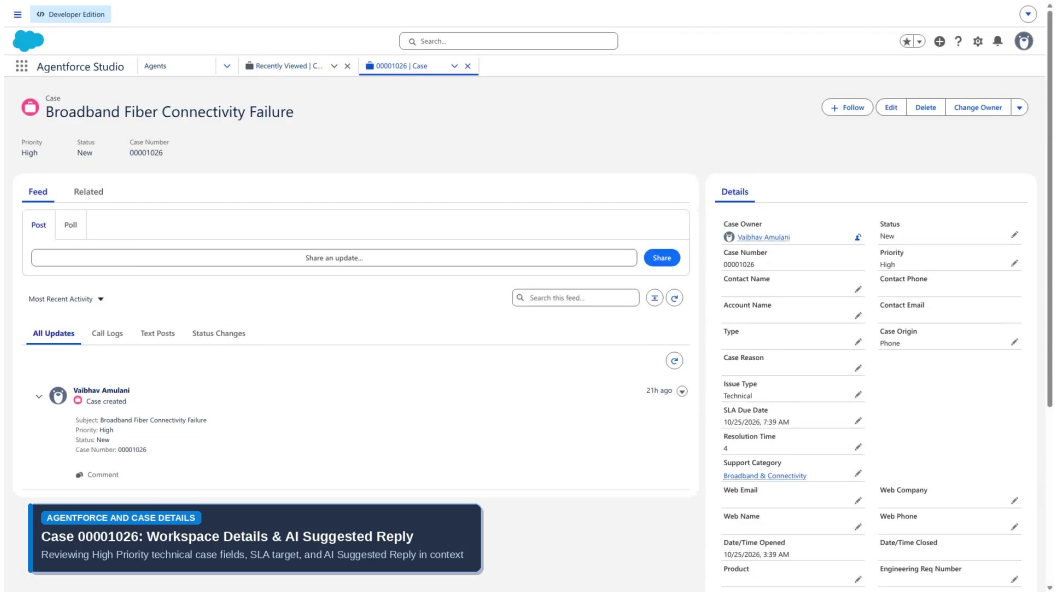


Figure 4. Case workspace and record details visible in the final demonstration (approximately 05:10).

> The supplied demo also shows the configured AI Feedback records and the SLA Breach & Resolution Time Analysis report. A separate, new-case creation walkthrough and a direct profile-permission walkthrough were not confirmed in the supplied timeline audit; these should be added to the final video if required by the assessment.

6. Reporting, Feedback, and Agentforce

Packaged reports

Report	Purpose
Open_Cases_by_Priority	Provides an operational view of open Cases grouped by priority.
SLA_Breach_and_Resolution_Time_Analysis	Provides SLA-related status classification and resolution-time analysis.

SLA report formula note

The SLA analysis report includes row-level formula CDF1 (“SLA Breach Status”). Its formula classifies records as No SLA Target, SLA Breached, Within SLA, or Within SLA (Active), depending on deadline and status. The formula is report-level logic, not a persistent Case field. The packaged formula labels should be treated as the source of truth when describing the report.

AI feedback loop

AI_Feedback__c records are intended to capture feedback associated with a Case, including suggestion type, status, and agent comments. Reviewing feedback trends can help identify suggestions that agents find useful or need to correct. The existence of feedback records alone does not establish that an automated model-training or retraining loop is implemented.

Agentforce verification result

> The live AI_Customer_Support_Assistant Version 1 conversation preview was tested with a broadband-outage scenario. The assistant returned a response saying it could not retrieve specific troubleshooting guidance and offered escalation to a live agent. The preview executed and returned a response, but successful custom action invocation and grounded knowledge retrieval were not demonstrated.

Therefore, describe Agentforce as configured and explored in the demo, not as a fully verified end-to-end troubleshooting solution. Test the relevant action, knowledge source, permissions, and response path before claiming the intended behavior works.

7. Development-Org Test Evidence

The following records and outcomes were reported in the supplied implementation report. They are environment-specific and are not recreated automatically when deploying metadata. This document generator did not independently replay these tests.

Record	Scenario / state	Reported outcome
Case 00001026	Broadband fiber outage; High; New	Validation negative test and before-save SLA/reply population reported successful; report classified overdue.
Case 00001027	International roaming billing dispute; Medium; Closed	12-hour target populated; resolution Flow reported executed; report classified within SLA.
Case 00001028	5G eSIM activation failure; Medium; Working	12-hour target and suggested reply populated; report classified active/within SLA.
Case 00001029	Regional cell tower outage; High; Escalated	Past SLA deadline retained for breach testing; escalation notification path reported triggered.
AF-00001	Feedback linked to Case 00001026	Helpful; Suggested Reply.
AF-00002	Feedback linked to Case 00001027	Helpful; Case Classification.

Recommended regression tests before submission/deployment

Test	Expected result to verify
High priority + blank SLA due date	Validation error appears unless automation validly populates the due date before save.
High priority + valid SLA due date	Case saves and SLA target/deadline match configured policy.
Medium and Low priority cases	Resolution target matches configured policy (12 and 24 hours respectively).
Case becomes Escalated	Correct alert path executes once and reaches the intended recipient.
Case transitions to Closed	Resolution notification executes as configured.
Support Agent vs Support Manager	Each profile can access only the intended objects and fields.
Agentforce action and knowledge retrieval	Configured action is invoked successfully and response is grounded in available support guidance.
Reports after case updates	Report status and grouping reflect the saved Case values and formula logic.

8. Deployment, Documentation, and Maintenance

Repository contents

The Salesforce DX source package includes custom object and field metadata, validation rules, Flows, email templates, workflow alerts, profiles, tabs, the Lightning application, reports, Agentforce-related metadata, a package manifest, and this documentation PDF.

Pre-deployment checklist

Check	Required action
Environment	Use a disposable development/test org and confirm required Salesforce features are available.
Package validation	Install Salesforce CLI, authenticate to the test org, and run the Salesforce DX deployment validation.
Email configuration	Confirm that alert recipients are valid and appropriate for the target org; verify email deliverability.
Access control	Review Support Agent and Support Manager object/field permissions and compatibility with target-org settings.
Automation	Verify Flow activation, entry conditions, field assignments, notification paths, and resolution behavior.
Agentforce	Verify feature availability, action access, knowledge sources, and successful invocation; do not assume the preview proves these.
Regression testing	Run the test matrix in Section 7 and retain evidence of actual outcomes.

Suggested deployment validation command

```
sf project deploy validate --manifest manifest/package.xml --target-org <your-org-alias>
```

This command is a recommended next step; it has not been executed as part of preparing this PDF. Resolve deployment errors and run functional tests before deploying to an intended environment.

Maintenance cadence

When	Review
After each release/change	Validate XML/package changes, run regression tests, inspect Flow activation and email alert configuration.
Regular operations review	Monitor open-case volume, SLA classifications, overdue cases, and feedback trends in reports.
When an incident occurs	Review Flow errors, validation failures, record history, and access permissions; document root cause and corrective action.
After corrective action	Repeat affected tests and confirm the report/notification outcome reflects the corrected behavior.

9. Conclusion and Known Limitations

The project package brings together standard Salesforce Case management, custom support/SLA/feedback data, record-triggered automation, validation, email notifications, operational reports, support profiles, and Agentforce-related configuration. The design is intended to reduce manual triage effort and make SLA status easier to inspect.

What the available evidence supports

• The package contains the documented Salesforce metadata components.
• The supplied implementation report describes development-org scenarios and test outcomes.
• The supplied demonstration video visibly walks through Cases, configuration, AI Feedback, reporting, and Agentforce-related screens.
• The documentation now includes an architecture view and genuine screenshots captured from that supplied video.

What remains unverified

• Salesforce CLI deployment validation to a target org.
• End-to-end Agentforce troubleshooting with successful custom action invocation and grounded knowledge retrieval.
• Actual permission enforcement for the target org's profiles and any permission sets.
• Replaying the reported test matrix independently against the current package.
• Continuous operational monitoring after deployment.

Final status: Prepared for review, not yet certified as successfully deployable. Complete target-org validation and regression testing before describing the package as production-ready.

Source and evidence note

This document was prepared from the supplied project documentation PDF, the supplied Salesforce DX package, the SkillWallet requirement screenshots, and screenshots extracted from the supplied final demo video. Reported live-org outcomes are labelled as reported; no unsupported test results or fabricated Salesforce screens have been added.